

Structure-aware fine-grained instance segmentation for fetal brain ultrasound images

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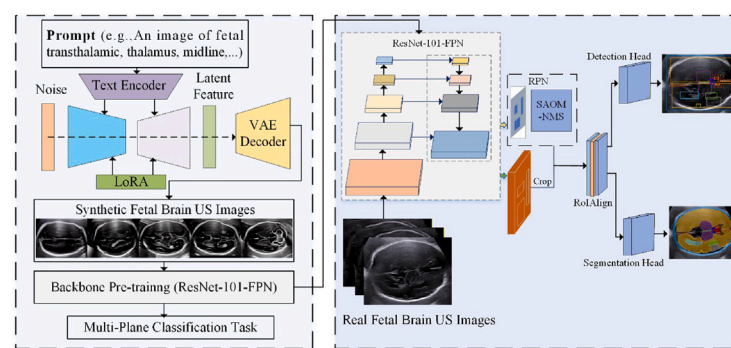
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HIGHLIGHTS

- A structure-aware instance segmentation framework is proposed for fetal brain ultrasound images across five standard planes.
- A structure-aware overlap metric and improved NMS algorithm are developed to enhance small structure segmentation performance.
- A fine-grained segmentation head and Laplacian regularization are designed to refine mask boundaries and preserve structural details.
- A domain-adaptive generative augmentation strategy is introduced to synthesize high-fidelity images for backbone pre-training.

GRAPHICAL ABSTRACT



The SAIS network architecture for fetal brain ultrasound segmentation. The left part illustrates the synthesis of fetal brain ultrasound images using stable diffusion with low-rank adaptation, which are subsequently used to pretrain the backbone network. The right part shows the fetal brain structure segmentation process using the backbone pretrained on the synthesized data.

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ABSTRACT

Fetal brain ultrasound segmentation remains challenging due to complex morphology and low-contrast imaging. Existing deep learning methods often lack anatomical awareness and spatial consistency, resulting in inaccurate boundaries, particularly for small structures, and they typically segment only a limited set of anatomical structures. To overcome these issues, a structure-aware fine-grained instance segmentation framework is proposed. The framework performs instance-level detection and mask prediction, enabling fine-grained segmentation of 25 anatomical structures across five standard planes. Specifically, a structure-aware overlap metric is proposed to mitigate the misclassification of small structures caused by conventional Intersection-over-Union based positive/negative sample assignment. Based on this metric, an improved non-maximum suppression algorithm is developed accordingly to enhance small-structure detection. A fine-grained segmentation head with an enlarged receptive field is designed to preserve essential details of anatomical structures. The proposed structure-aware Laplacian regularization strategy further refines the spatial consistency of masks and boundary accuracy through graph Laplacian smoothing. Additionally, a domain-adaptive generative augmentation strategy based on Stable

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Diffusion with Low-Rank Adaptation is applied to synthesize anatomically diverse fetal brain ultrasound images and improve segmentation model generalization. Extensive experiments demonstrate that the proposed framework consistently outperforms existing baselines on fetal brain ultrasound segmentation.

1. Introduction

The fetal brain represents the central component of the nervous system, whose maturation not only determines neural functional integrity in utero but also significantly influences postnatal cognitive abilities [1–3]. In prenatal assessment, accurate segmentation of key fetal brain anatomical structures is essential for reliable measurement of growth parameters and quantitative evaluation of developmental trajectories, thereby facilitating early detection of neurological abnormalities and informing clinical decision-making [4,5].

Ultrasound imaging, characterized by its real-time capability and operational convenience, has become the preferred modality for prenatal brain screening. The guidelines of the International Society of Ultrasound in Obstetrics and Gynecology [6,7] explicitly recommend ultrasound as the primary examination for fetal neurodevelopmental evaluation. However, the diagnostic accuracy of ultrasound is highly dependent on the precise structural segmentation in standard fetal brain planes, such as the cerebellum, lateral ventricles, and thalamus, as these segmentation results directly impact the reliability of subsequent morphological and developmental assessments.

A major challenge in fetal brain structure segmentation stems from the inherent limitations of ultrasound imaging, which is frequently corrupted by speckle noise, artifacts, and fetal motion. These factors reduce image contrast and blur structural boundaries, substantially complicating segmentation. Conventional segmentation approaches are highly operator-dependent, introducing considerable subjectivity and poor reproducibility, thereby limiting the reliability of fetal brain assessment [8,9].

Deep learning models, particularly convolutional neural networks (CNNs), have demonstrated exceptional performance in medical image segmentation and offer a promising solution for fetal brain structure segmentation [10–13]. By leveraging end-to-end training and hierarchical feature learning, existing methods have successfully segmented key anatomical structures, including the cavum septum pellucidum, lateral ventricles, and skull [14,15]. Nevertheless, most current approaches employ semantic segmentation frameworks, which, despite their pixel-wise prediction capability, are unable to distinguish multiple instances of the same anatomical category.

Despite the success of deep learning in improving segmentation accuracy, fetal brain segmentation still faces several critical challenges. The brain's morphology changes dynamically across gestational stages, limiting model generalization across different ages. Existing methods typically fail to incorporate anatomical relationships and spatial consistency, often producing inaccurate boundaries. Furthermore, most approaches focus only on a limited set of key structures, failing to achieve comprehensive whole-brain, multi-structure segmentation and demonstrating insufficient performance on small-scale anatomical regions. These limitations hinder the clinical application of automated fetal brain segmentation techniques.

To address these challenges, we propose a structure-aware instance segmentation framework (SAIS) for fetal brain ultrasound images, inspired by the Mask R-CNN architecture [16]. SAIS performs independent detection and pixel-level mask prediction for each anatomical structure, enabling differentiation of individual instances within the same category and achieving superior performance in fine-grained segmentation of complex brain morphologies. It can segment 25 anatomical structures across five standard planes, including the transcalvarial plane (TCV), transcavum septi pellucidi plane (TCSP), transthalamic plane (TTA), transventricular plane (TVL), and transcerebellar plane (TCB).

Specifically, to address the limitation of conventional methods that rely solely on the Intersection-over-Union (IoU) between two bounding

boxes for positive/negative sample assignment—often misclassifying small structures as negative samples—we propose a structure-aware overlap metric (SAOM), which extends IoU by incorporating the ratio between the intersection area and the ground-truth or high-confidence candidate bounding box area. Based on SAOM, we further develop an improved non-maximum suppression algorithm (SAOM-NMS), significantly enhancing the detection and segmentation of small fetal brain structures. To alleviate feature compression in the original segmentation head, we design a fine-grained segmentation head (FGSH) with an enlarged receptive field, preserving essential anatomical information and subtle structural details. Additionally, we propose a structure-aware Laplacian regularization (SALR) strategy, modeling each region-of-interest (RoI) mask as a graph signal and refining it through graph Laplacian smoothing. This reduces noise and discontinuities, improving spatial consistency and boundary accuracy of the predicted masks. Finally, to improve the generalization of fetal brain ultrasound segmentation under dynamic anatomical changes, we incorporate a domain-adaptive generative augmentation strategy (DAGA) using Stable Diffusion with Low-Rank Adaptation (LoRA) [17,18]. DAGA synthesizes diverse, high-fidelity fetal brain ultrasound images capturing continuous structural variations, which are then used to pre-train the backbone network of SAIS.

The major contributions are summarized as follows:

- A structure-aware instance segmentation framework (SAIS) is proposed for fetal brain ultrasound images, enabling segmentation of 25 anatomical structures across five standard planes, i.e., TCV, TCSP, TTA, TVL, and TCB.
- A structure-aware overlap metric (SAOM) is proposed, and an improved non-maximum suppression algorithm (SAOM-NMS) is developed to mitigate the misclassification of small structures during sample assignment, thereby improving the detection and segmentation performance for small anatomical targets.
- A fine-grained segmentation head (FGSH) is designed to enlarge the receptive field, and a structure-aware Laplacian regularization (SALR) strategy is formulated to refine RoI mask boundaries, enhancing structural detail preservation and spatial consistency.
- A domain-adaptive generative augmentation strategy (DAGA) based on Stable Diffusion with Low-Rank Adaptation (LoRA) is introduced to synthesize high-fidelity fetal brain ultrasound images for backbone pre-training, improving the generalization capability of SAIS.

2. Related work

2.1. Semantic segmentation in fetal brain ultrasound

Deep learning-based automatic segmentation methods for fetal brain structures have been developed [10–13]. Most existing studies formulate the task as semantic segmentation on 2D or 3D ultrasound images, aiming to extract key anatomical regions such as the ventricular system, corpus callosum, and cerebellum.

Early and representative works improved encoder-decoder backbones by enhancing feature selection and localization. Zeng et al. [14] integrated an attention-gated module into the V-Net framework [19], improving the precision of fetal head measurements. Namburete et al. [20] developed a multi-task 3D CNN for joint recognition and segmentation of multiple fetal brain structures. Beyond pure segmentation, several studies emphasized region localization and feature transfer to handle ultrasound appearance variations: Kwitt et al. [15] localized key anatomical regions using dynamic texture templates, while Chen et al. [21] applied transfer learning to adapt features from natural images

to fetal ultrasound, improving the robustness of plane recognition. Similarly, Oghli et al. [22] proposed the MFP-Unet model, automatically detecting salient regions to achieve precise fetal skull segmentation.

With further advancements, stronger spatial context modeling was introduced. 3D CNNs were explored to better capture volumetric contextual cues and improve boundary coherence; for example, Hesse et al. [23] proposed a 3D CNN for simultaneous segmentation of multiple regions such as the choroid plexus, cerebellum, and cavum septi pellucidi. To enhance robustness and multi-scale feature representation, Szentimrey et al. [24] designed a multi-branch U-Net ensemble that fuses multi-scale features. To mitigate noise and blurring in ultrasound images, Qiao et al. [25] employed dilated convolutions to enlarge receptive fields and improve recognition in complex scenarios. Attention mechanisms have also proven effective in focusing on salient anatomical cues. Shu et al. [26] introduced channel attention, while Yang et al. [27] proposed a hybrid spatial-channel attention network to reduce errors caused by fetal pose variations.

Given the morphological diversity of fetal brain structures across gestational ages, some studies further incorporated anatomical priors and stage-aware strategies to improve generalization. Karimi et al. [28] combined multi-atlas fusion and deformable registration for cross-gestational segmentation, and Mohana et al. [29] developed a multi-strategy fusion approach for gestational-stage-specific head circumference estimation. Recently, data-driven paradigms have expanded with large-scale datasets and prompt-based ideas. Alzubaidi et al. [30] constructed a high-resolution fetal head ultrasound dataset and proposed FetSAM [31], introducing prompt learning into segmentation tasks. Singh et al. [32] combined U-Net++ and ResNet architectures to design an attention-enhanced framework and incorporated Grad-CAM++ for interpretable visualization. To address imperfect data characteristics in complex medical image analysis, a local-global attention fusion framework with tensor decomposition was proposed [33]. The method integrates CNNs and Transformers to jointly model local and global contextual features and to obtain compact feature representations. To improve small-sized object detection, an atrous spatial pyramid pooling balanced feature pyramid network was introduced [34]. This approach enhances multi-scale feature fusion using atrous convolutions and a balanced module, thereby preserving fine-grained spatial details.

Despite notable progress, semantic segmentation-based pipelines still face practical limitations in fine-grained fetal brain ultrasound analysis. Many approaches focus on a limited set of key structures, while comprehensive whole-brain, multi-structure segmentation remains relatively scarce. Moreover, the coexistence of large and tiny anatomical regions makes learning highly imbalanced, and conventional training strategies may suppress small structures, leading to missed or fragmented predictions. In addition, ultrasound-specific speckle noise, low contrast, and ambiguous boundaries demand stronger instance-level separation and spatial consistency modeling than what conventional semantic segmentation typically provides.

These limitations naturally motivate considering an instance segmentation formulation, where detection and mask prediction are decoupled and each anatomical structure can be handled as an individual instance.

2.2. Instance segmentation

Unlike semantic segmentation, instance segmentation aims to detect and segment individual objects, providing both a bounding box and a pixel-wise mask. Making it well-suited for fine-grained multi-structure analysis [35]. Mask R-CNN [16] established the foundational framework for this paradigm by adding a mask prediction branch to the Faster R-CNN detector. Beyond Mask R-CNN, numerous alternative models have been proposed. PANet [36] introduced path aggregation and adaptive pooling to strengthen multi-scale feature interactions. SOLO [37] treated instance segmentation as a location-guided pixel classification task, avoiding proposal generation while maintaining high accuracy.

CondInst [38] generated instance-dependent convolution kernels using conditional convolutions, enabling flexible mask prediction. BlendMask [39] enhanced Mask R-CNN by adding a specialized blending module that fused global and local information. Even real-time detectors such as the YOLO series (e.g., YOLOv11 [40]) have evolved to support instance segmentation, balancing speed and accuracy. MEInst [41] encodes instance masks into compact low-dimensional vectors and directly regresses them within a single-stage detector, enabling efficient instance segmentation without pixel-level mask prediction branches.

Although these generic instance segmentation models have shown great promise in computer vision, they are not directly transferable to fetal brain ultrasound image segmentation. A primary challenge is that these approaches rely on Intersection-over-Union (IoU) for sample assignment, which is overly sensitive for small fetal brain structures—slight deviations can lead to these structures being discarded as background. Additionally, generic mask heads typically lack the regularization needed to handle the fuzzy, discontinuous boundaries characteristic of ultrasound images. Motivated by these gaps, our approach is primarily inspired by the instance segmentation framework, and is built upon Mask R-CNN [16]. In particular, we address the small structure segmentation issue, enhance fine-grained mask prediction, and improve spatial consistency and boundary accuracy. Additionally, to better generalize under dynamic anatomical changes, we incorporate a domain-adaptive generative augmentation strategy using Stable Diffusion with LoRA [17,18].

3. Method

3.1. Overall architecture

The SAIS network architecture is illustrated in Fig. 1, it is designed to segment 25 fine-grained anatomical structures across five fetal brain ultrasound standard planes, i.e., TCV, TCSP, TTA, TVL, and TCB. Built upon Mask R-CNN, SAIS begins with a generative initialization phase to enhance feature representation. A domain-adaptive generative augmentation strategy based on Stable Diffusion with Low-Rank Adaptation is employed to synthesize high-fidelity ultrasound images for all five planes. These synthetic images are used to pre-train the ResNet-101-FPN backbone through a supervised standard fetal brain ultrasound plane classification task, enabling the backbone to learn plane-specific texture patterns and global anatomical features prior to segmentation. After feature extraction, a novel structure-aware overlap metric is integrated into the Region Proposal Network (RPN) to prevent small but clinically important targets from being misclassified as background. The SAOM-based non-maximum suppression algorithm is further applied to filter redundant proposal bounding boxes while preserving valid ones. In the final instance segmentation stage, RoI-aligned features for each Region of Interest are processed through two parallel branches. The first branch performs classification and bounding-box regression, and the second branch is a fine-grained segmentation head with an enlarged receptive field to capture broader contextual information and preserve essential structural details. To further improve segmentation quality, a structure-aware Laplacian regularization strategy is proposed, modeling the predicted mask as a graph signal and performing Laplacian smoothing to reduce noise and spatial discontinuities. Together, these components enable SAIS to produce precise, pixel-level, and anatomically coherent segmentation results across all five standard fetal brain ultrasound planes.

3.2. Domain-adaptive generative augmentation strategy

To address the limited generalization of fetal brain ultrasound segmentation models, we introduce a domain-adaptive generative augmentation method for fetal brain ultrasound image synthesis across five clinically standardized planes. The main purpose of the domain-adaptive strategy is to synthesize fetal brain ultrasound images whose texture and appearance are better matched to the real ultrasound domain, so that the

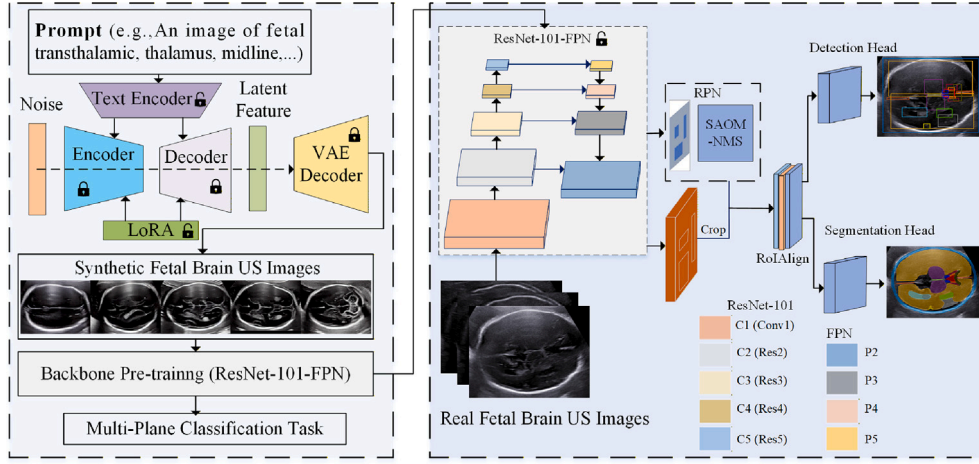


Fig. 1. Overview of the proposed framework. The left panel shows the synthesis of fetal brain ultrasound images using Stable Diffusion with Low-Rank Adaptation, guided by text prompts. The generated images are then used to pre-train the backbone (ResNet-101-FPN configuration as Mask R-CNN) through a five-class classification task corresponding to the five fetal brain standard planes, producing pretrained backbone weights. The right panel presents the downstream fetal brain structure segmentation pipeline on real ultrasound images. The segmentation network is initialized with the backbone pretrained on the synthesized data, and then trained end-to-end with our overall segmentation architecture.

synthesized data can be used to pre-train the segmentation backbone and improve generalization. In other words, the adaptation primarily targets cross-subject/cross-case variability in real clinical scans e.g., noise, contrast, attenuation and acoustic shadows.

The input $x \in \mathbb{R}^{H \times W \times 3}$ represents a raw fetal brain ultrasound image, and y is the corresponding anatomical prompt (e.g., “transthalamic plane, thalamus, midline”). The method is built upon the Stable Diffusion v1.5 architecture [17]. As shown in Fig. 2, during training, the input image x is first compressed by a pre-trained Variational Autoencoder (VAE) into a latent representation $z_0 = \mathcal{E}(x)$, which is then progressively corrupted by Gaussian noise ϵ to produce a noisy latent z_t . Simultaneously, the text prompt y is processed by a CLIP Text Encoder [42] to generate semantic embeddings $\tau_\theta(y)$. The core generative network, a U-Net backbone ϵ_θ , takes both the noisy latent z_t and the text embeddings $\tau_\theta(y)$ as inputs. The text embedding guides the model to synthesize specific fetal brain structures. The optimization objective is to minimize the denoising error $L = \|\epsilon - \epsilon_\theta(z_t, t, \tau_\theta(y))\|^2$, forcing the model to reconstruct the anatomical features of x conditioned on the semantic description y .

To efficiently adapt the pre-trained model to the unique texture of fetal brain ultrasound images, we employ the LoRA strategy [18]. We freeze the original U-Net weights W_0 and inject trainable rank-decomposition matrices into the attention layers, defined as $\Delta W = \frac{\alpha}{r} BA$. Uniquely, we configure the network rank to a high value of $r = 64$ (with scaling factor $\alpha = 32$).

Our training strategy relies on structured anatomical prompts to encode standard fetal brain planes (e.g., transcalvarial, transcavum septi pellucidi, transthalamic, transventricular, and transcerebellar planes). By enumerating critical landmarks like the thalamus and cerebellar vermis, these prompts explicitly condition the model for anatomical fidelity. Training parameters included an image resolution of 512×512 , a batch size of 1, and the AdamW optimizer (U-Net learning rate: 1×10^{-4} ; Text Encoder learning rate: 5×10^{-5}). The model training was performed for 20 epochs. Additionally, a Noise Offset of 0.1 was applied to the noise schedule to ensure the synthesis of realistic acoustic shadows and high-contrast clinical appearances.

The inference was performed using 30 denoising steps with a Classifier-Free Guidance scale of 7.0 to ensure structural fidelity, and the LoRA weight scale is 0.7. For each fetal brain ultrasound standard plane, we generated 5000 synthetic images to construct a plane-specific augmented dataset. The synthesized images were subsequently employed to train the downstream backbone, ResNet-101-FPN. It was reconfigured as a five-category classification model by attaching a fully connected layer to its final feature stage. The five output categories correspond to the standard fetal brain ultrasound planes: transcalvarial, transcavum septi pellucidi, transthalamic, transventricular, and transcerebellar.

3.3. Structure-aware overlap metric and improved NMS

The RPN generates a large number of candidate bounding boxes, a substantial portion of which are negative samples that do not contain key

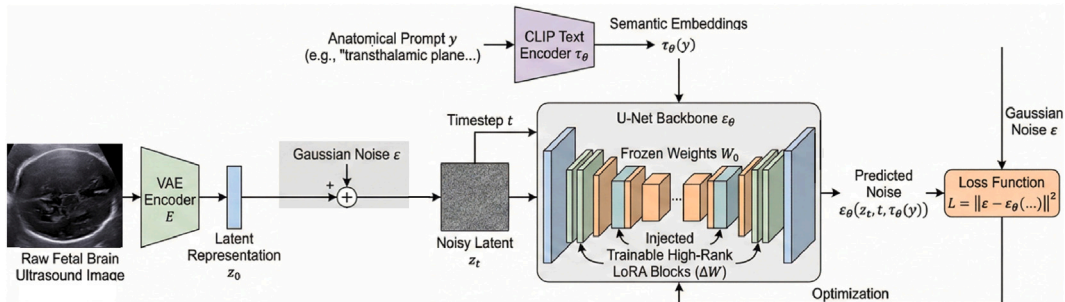


Fig. 2. The domain-adaptive generative augmentation model architecture.

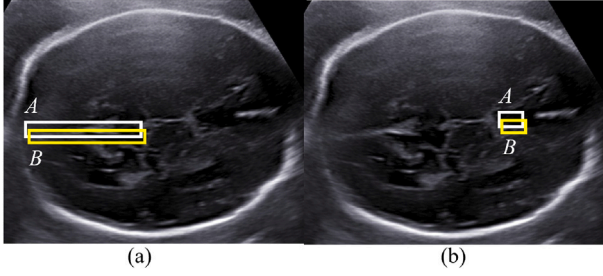


Fig. 3. The candidate bounding boxes corresponding to elongated or small anatomical structures.

brain structures. To enable the model to learn which candidate bounding boxes correspond to true targets, the IoU between each candidate bounding box and the ground-truth annotation is calculated, and samples are classified as positive or negative based on a predefined threshold. While this approach is effective for detecting large brain anatomical structures, it is highly sensitive to small or elongated structures. For such structures, the union area between a candidate bounding box and the ground-truth bounding box is relatively large compared to their intersection, resulting in a significantly reduced IoU. Consequently, these candidate bounding boxes are frequently misclassified as negative samples, leading to decreased recall for small or slender brain structures. As illustrated in Fig. 3, A and B denote the ground-truth bounding box and the RPN-generated candidate bounding box, respectively. In Fig. 3(a), the fetal falx cerebri is a slender structure; due to the large union and small intersection, candidate bounding box B is erroneously classified as negative. Similarly, in Fig. 3(b), the smaller column of the fornix structure's candidate bounding box B is prone to misclassification due to a low IoU.

To address this issue, we propose a structure-aware overlap metric for evaluating the overlap between two candidate bounding boxes, which considers not only the standard IoU but also the proportion of their intersection area relative to the area of the ground-truth bounding box (or a high-confidence candidate bounding box A):

$$\text{IoUA} = \lambda \cdot \frac{\text{area}(A \cap B)}{\text{area}(A \cup B)} + (1 - \lambda) \cdot \frac{\text{area}(A \cap B)}{\text{area}(A)}, \quad (1)$$

where A and B denote candidate bounding boxes. When used for positive-negative sample assignment, A represents the ground-truth annotation, while B corresponds to a candidate bounding box with a lower classification confidence score than A , the λ is set to 0.8.

Based on this metric, we developed the SAOM-NMS algorithm. By incorporating the weighted IoUA, the NMS can more intelligently suppress redundant bounding boxes, prioritizing detections with high confidence scores and accurate localization. Specifically, let the set of candidate bounding boxes for fetal brain anatomical structures be:

$$B = (b_i, s_i)_{i=1}^N, \quad (2)$$

where $b_i \in \mathbb{R}^4$ represents the coordinates of the i -th bounding box (typically the top-left and bottom-right corners), and $s_i \in [0, 1]$ denotes the corresponding confidence score.

The SAOM-NMS first sorts all candidate bounding boxes in descending order according to their confidence scores:

$$B \leftarrow \text{sort}(B, \text{key} = s_i, \text{descending}) \quad (3)$$

It then initializes the set of retained bounding boxes $\mathcal{R} = \emptyset$ and iteratively selects the highest-scoring bounding box $b_{\max}$, adding it to $\mathcal{R}$:

$$\mathcal{R} \leftarrow \mathcal{R} \cup b_{\max} \quad (4)$$

For each remaining candidate bounding box $b_j \in B \setminus b_{\max}$, the SAOM with $b_{\max}$ is computed as:

$$\text{IoUA}(b_{\max}, b_j) = \lambda \cdot \frac{\text{area}(b_{\max} \cap b_j)}{\text{area}(b_{\max} \cup b_j)} + (1 - \lambda) \cdot \frac{\text{area}(b_{\max} \cap b_j)}{\text{area}(b_{\max})} \quad (5)$$

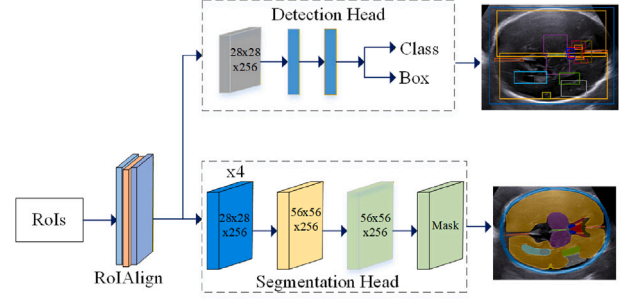


Fig. 4. Architecture of the refined segmentation head model for fetal central nervous system structures.

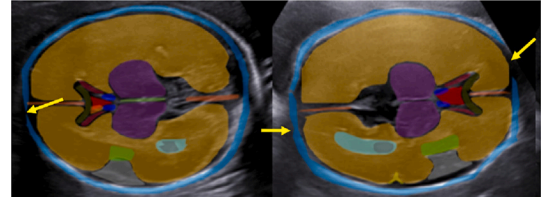


Fig. 5. Discontinuity in the segmented skull.

All candidate bounding boxes whose IoUA with $b_{\max}$ exceeds the threshold $\theta \in [0, 1]$ are then removed:

$$B \leftarrow (b_j, s_j) \in B \mid \text{IoUA}(b_{\max}, b_j) < \theta \quad (6)$$

This process iterates until B is empty, and the final set of retained bounding boxes is given by:

$$\mathcal{R} = \text{NMS}(B, \theta) \quad (7)$$

The threshold θ controls the suppression strength for overlapping bounding boxes and is set to 0.7 in our experiments.

3.4. Fine-grained segmentation head network

The segmentation head of Mask R-CNN excessively compresses features, making it challenging to segment ring-shaped structures, such as the skull. In this work, we propose a fine-grained segmentation head for brain anatomical structures with an enlarged receptive field.

As illustrated in Fig. 4, the RoIs generated by the RPN are first fed into RoIAlign. RoIAlign divides each RoI into a fixed number of bins (7×7) and samples multiple points within each bin. The features of these sampling points are obtained via bilinear interpolation, producing aligned feature maps of size 28×28 . These feature maps are then processed by four convolutional layers with a kernel size of 3×3 , a stride of 1, and padding of 1, resulting in output feature maps of size $28 \times 28 \times 256$. Next, the feature maps are passed through a transposed convolution layer with a kernel size of 2×2 and a stride of 2, producing feature maps of size $56 \times 56 \times 256$. Additional convolutional layers further refine the feature maps, maintaining the same size of $56 \times 56 \times 256$. Finally, a convolution layer generates the refined segmentation mask for each RoI of fetal brain anatomical structures.

3.5. Structure-aware laplacian regularization

The segmentation head predicted masks of brain structures often exhibit issues such as the blurred boundaries, structural discontinuities, or local inaccuracies, as indicated by the yellow arrows in Fig. 5. To address this limitation, we propose a structure-aware Laplacian regularization. The core idea of this method is to treat the mask prediction of each RoI as a graph signal and construct a local pixel graph structure. By

leveraging the Laplacian smoothing theory in graph signal processing, the proposed approach effectively suppresses noise and discontinuities within the mask, thereby enhancing the model's structural awareness.

Specifically, for each RoI feature map obtained after the RoIAlign:

$$\mathcal{F} = \begin{bmatrix} f_1^\top \\ f_2^\top \\ \vdots \\ f_N^\top \end{bmatrix} \in \mathbb{R}^{N \times c}, \quad N = h \times w, \quad (8)$$

where c , h , and w denote the number of channels, height, and width of the feature map, respectively. Each $f_i \in \mathbb{R}^c$ represents the feature vector of the i -th pixel, corresponding to its spatial coordinate $p_i \in \mathbb{R}^2$. Based on this, a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ is constructed, where each node $v_i \in \mathcal{V}$ represents a pixel. The edge weights (i.e., adjacency matrix) $\mathbf{W} \in \mathbb{R}^{N \times N}$ of the graph are defined as:

$$W_{ij} = \begin{cases} \exp\left(-\frac{\|f_i - f_j\|_2^2}{\sigma_f^2} - \frac{\|p_i - p_j\|_2^2}{\sigma_p^2}\right), & \text{if } \|p_i - p_j\|_2 \leq r, \\ 0, & \text{otherwise.} \end{cases} \quad (9)$$

Here, σ_f and σ_p control the decay rates of feature similarity and spatial distance, respectively (corresponding to the median of feature distances and a standard spatial distance value). The parameter r denotes the perceptual neighborhood radius (corresponding to an 8-neighborhood in this work). Only local neighboring pixels are connected to maintain the sparsity of the graph structure. The adjacency matrix $\mathbf{W} \in \mathbb{R}^{N \times N}$ thus jointly encodes both semantic consistency and spatial connectivity. The term $\|f_i - f_j\|_2^2$ reflects the deep feature distance between two pixels, ensuring that semantically consistent regions are closely connected in the graph, while $\|p_i - p_j\|_2^2$ ensures that connections are only established between spatially adjacent pixels, thereby preserving local spatial consistency.

The degree diagonal matrix of the graph $\mathbf{D} \in \mathbb{R}^{N \times N}$ is defined as:

$$D_{ii} = \sum_{j=1}^N W_{ij}, \quad (10)$$

then the unnormalized graph Laplacian matrix is obtained as:

$$\mathbf{L} = \mathbf{D} - \mathbf{W}. \quad (11)$$

Each RoI probability mask predicted by the segmentation head, $\hat{\mathbf{m}} \in [0, 1]^{1 \times h \times w}$, is flattened into a vector:

$$\hat{\mathbf{m}} = \begin{bmatrix} \hat{m}_1 \\ \hat{m}_2 \\ \vdots \\ \hat{m}_N \end{bmatrix} \in \mathbb{R}^N. \quad (12)$$

Based on the above graph structure, the Laplacian regularization term is defined as:

$$L_{\text{lap}} = \hat{\mathbf{m}}^\top \mathbf{L} \hat{\mathbf{m}} = \hat{\mathbf{m}}^\top (\mathbf{D} - \mathbf{W}) \hat{\mathbf{m}} = \hat{\mathbf{m}}^\top \mathbf{D} \hat{\mathbf{m}} - \hat{\mathbf{m}}^\top \mathbf{W} \hat{\mathbf{m}} \quad (13)$$

Since $\mathbf{D}$ is a diagonal matrix, it can be expanded as:

$$\hat{\mathbf{m}}^\top \mathbf{D} \hat{\mathbf{m}} = \sum_{i=1}^N D_{ii} \hat{m}_i^2 = \sum_{i=1}^N \sum_{j=1}^N W_{ij} \hat{m}_i^2, \quad (14)$$

and

$$\hat{\mathbf{m}}^\top \mathbf{W} \hat{\mathbf{m}} = \sum_{i=1}^N \sum_{j=1}^N \hat{m}_i W_{ij} \hat{m}_j. \quad (15)$$

Substituting these into the expression yields:

$$L_{\text{lap}} = \sum_{i,j} W_{ij} \hat{m}_i^2 - \sum_{i,j} W_{ij} \hat{m}_i \hat{m}_j. \quad (16)$$

Since the adjacency matrix $\mathbf{W}$ is symmetric, i.e., $W_{ij} = W_{ji}$, consider:

$$\sum_{i,j} W_{ij} (\hat{m}_i - \hat{m}_j)^2 = \sum_{i,j} W_{ij} (\hat{m}_i^2 - 2\hat{m}_i \hat{m}_j + \hat{m}_j^2). \quad (17)$$

Using symmetry, we have:

$$\sum_{i,j} W_{ij} \hat{m}_i^2 = \sum_{i,j} W_{ij} \hat{m}_j^2, \quad (18)$$

and the above can be simplified to:

$$\sum_{i,j} W_{ij} (\hat{m}_i - \hat{m}_j)^2 = 2 \sum_{i,j} W_{ij} \hat{m}_i^2 - 2 \sum_{i,j} W_{ij} \hat{m}_i \hat{m}_j. \quad (19)$$

Combining with the previous expression, we obtain:

$$L_{\text{lap}} = \frac{1}{2} \sum_{i,j} W_{ij} (\hat{m}_i - \hat{m}_j)^2. \quad (20)$$

Since $W_{ij} = 0$ for non-adjacent nodes, the above can be rewritten as a summation over all edges $\mathcal{E}$:

$$L_{\text{lap}} = \sum_{(i,j) \in \mathcal{E}} W_{ij} (\hat{m}_i - \hat{m}_j)^2. \quad (21)$$

This regularization term enforces that if two pixels of the fetal brain anatomical structure are close in both the feature space and physical space, the difference between their predicted mask values should be minimized. This ensures that pixels with similar anatomical structures yield similar predictions, thereby enhancing the spatial and semantic consistency of the mask predictions and improving the accuracy and completeness of boundaries.

The total loss function is defined as

$$L_{\text{total}} = L_{\text{cls}} + L_{\text{box}} + L_{\text{mask}} + \lambda \cdot L_{\text{lap}}, \quad (22)$$

where λ is set to 0.001. Here, L_{cls} , L_{box} , and L_{mask} represent the classification, bounding box regression, and anatomical structure segmentation losses for fetal brain structures, respectively, following the same definitions as in Mask R-CNN.

4. Implementation and results

4.1. Dataset and preprocessing

The fetal brain ultrasound data used in this study were collected from the Affiliated Shenzhen Maternity & Child Healthcare Hospital of Southern Medical University, covering routine examinations of 2600 pregnant women (gestational age 18–38 weeks) from May 2020 to March 2022. A total of 12,000 images across five standard transverse planes were acquired and annotated for deep learning model training and validation. Specifically, 2000 images corresponded to the transcalvarial plane, 2200 images to the transcavum septi pellucidi plane, and 2600 images each to the transthalamic plane, transventricular plane, and transcerebellar plane. The dataset was randomly divided into training, validation, and test sets with a ratio of 7:2:1. Acquisition was performed using Philips, GE, Mindray, Kaile, and Samsung ultrasound devices with probe frequencies ranging from 1 to 8 MHz.

For anatomical structure segmentation, as shown in the Table 1, we labeled 25 anatomical structures across five standard planes, including the transcalvarial, transcavum septi pellucidi, transthalamic, transventricular, and transcerebellar planes. Boundaries in each fetal brain ultrasound image were manually delineated by physicians using software developed by our research group. Initial annotations were performed by a doctor with 1–3 years of experience in fetal brain ultrasound, and subsequently reviewed by two senior prenatal ultrasound experts with 15 and 23 years of experience, respectively. Images with inconsistent review results were excluded. This study was approved by the institutional ethics committee.

4.2. Experimental setup

The model was implemented under the Detectron2 framework [43]. Input images were normalized, with the side resized to 512 pixels, and were single-channel (grayscale ultrasound images). Training data were augmented using random horizontal flipping and multi-scale resizing to improve model generalization.

The backbone network was ResNet-101-FPN. Training used a batch size of 3 for a maximum of 200,000 iterations, with an initial learning rate of 0.001, optimized using Adam with a weight decay of 0.0001, the random seed is -1 . The RPN follows the default Mask R-CNN configuration in Detectron2. Anchors are generated on FPN levels with sizes of [32, 64, 128, 256, 512] and aspect ratios [0.5, 1.0, 2.0]. The batch size per image is 256. The RPN losses include binary cross-entropy for objectness and Smooth-L1 for box regression. Training was performed in two stages, with the learning rate reduced to one-tenth of the initial value after 10,000 steps in the first stage. All experiments were conducted on an NVIDIA P100 Pascal GPU with 16 GB memory, keeping annotation information and other training hyperparameters (e.g., Detectron2 [43] defaults) unchanged to ensure comparability and reproducibility.

4.3. Evaluation metrics

To objectively evaluate the proposed method for detecting and segmenting fetal brain anatomical structures, standard COCO metrics [44] were used, including mAP, AP_{50} , and AP_{75} . The mean Average Precision (mAP) is computed as the average of AP values over 10 different IoU thresholds ranging from 0.5 to 0.95 with a step of 0.05. AP is defined as the average precision at recall levels from 0.1 to 1, with the IoU threshold determining whether a predicted bounding box or segmentation mask is correct. AP_{50} and AP_{75} correspond to AP at IoU thresholds of 0.5 and 0.75, respectively. mAP is calculated as:

$$mAP = \frac{\sum_{IoU \in [0.50, 0.95]} AP_{IoU}}{10} \quad (23)$$

where

$$AP = \frac{\sum_{r \in \{0.0, 0.1, \dots, 1\}} \max_{r' \geq r} p(r')}{11} \quad (24)$$

$$IoU = \frac{\text{area}(B_p \cap B_{gt})}{\text{area}(B_p \cup B_{gt})} \quad (25)$$

Here, B_p and B_{gt} denote the predicted and ground truth bounding boxes, respectively.

Table 1
List of abbreviations for fetal brain structures.

No.	Abbr.	Full Name	No.	Abbr.	Full Name
1	SK	Skull	14	CF	Columna Fornicis
2	FC	Falx Cerebri	15	POS	Parieto-Occipital Sulcus
3	CP	Cerebral Parenchyma	16	CC	Callosum
4	TH	Thalami	17	INS	Insula
5	TV	Third Ventricle	18	IFS	Inferior Frontal Sulcus
6	PHLV	Posterior Horn of Lateral Ventricle	19	CH	Cerebellar Hemispheres
7	AHLV	Anterior Horn of Lateral Ventricle	20	CV	Cerebellar Vermis
8	ChP	Choroid Plexus	21	MB	Midbrain
9	SF	Sylvian Fissure	22	TC	Tentorium Cerebelli
10	CSP	Cavum Septi Pellucidi	23	AM	Aqueduct of Midbrain
11	SP	Septum Pellucidum	24	CM	Cisterna Magna
12	STS	Superior Temporal Sulcus	25	CMF	Cerebellomedullary Fissure
13	ITS	Inferior Temporal Sulcus			

4.4. Detection and segmentation results

4.4.1. Segmentation results

The columns 3–8 report segmentation results in Table 2 for Mask R-CNN and the proposed method SAIS on 25 fetal brain anatomical structures in the test dataset, the statistical significance is assessed using paired Wilcoxon signed-rank tests. The best average results in the table are highlighted in bold. According to the mean values of mAP, AP_{50} , and AP_{75} , the proposed method outperforms Mask R-CNN significantly, with improvements of 5.10%, 4.41%, and 4.45%, respectively.

Among large structures, such as the skull, thalami, cerebral parenchyma, and cerebellar hemispheres, the proposed method demonstrates superior boundary adherence. Notably, the skull segmentation mAP increases dramatically by 9.52%, with AP_{50} improving by 9.07% and AP_{75} by 9.38%. This indicates that the proposed method effectively handles the complex curvature and acoustic shadows often associated with the skull, which are challenging for the baseline method (52.45% mAP). Other large structures also show steady improvements; for example, the cerebellar hemispheres and thalami gain 2.28% and 1.08% in mAP, respectively.

For small, thin, or structurally complex regions, the segmentation improvements are even more pronounced. The anterior horn of the lateral ventricle achieves a remarkable increase in mAP of 19.89% (from 33.89% to 53.78%), with AP_{75} surging by 14.60%. Similarly, the parieto-occipital sulcus sees an mAP increase of 9.51% and an AP_{50} increase of 7.20%. The tentorium cerebelli also improves by 7.05% in mAP and 6.87% in AP_{75} . These results indicate that the proposed fine-grained segmentation head and structure-aware Laplacian regularization strategy are highly effective in preserving the geometric details of challenging anatomical structures.

Fig. 6 shows the segmentation results of Mask R-CNN and the proposed method on fetal brain ultrasound images. The first row shows the original images (OI), the second row provides the ground-truth annotations (GT), and the third and fourth rows display the segmentation results of the proposed model and Mask R-CNN, respectively. Figures (a)–(e) correspond to the TCV, TCSP, TTA, TVL, and TCB planes. As illustrated, the proposed fine segmentation network markedly enhances segmentation performance. In particular, it effectively mitigates the degradation caused by excessive feature compression and achieves more accurate delineation of skull boundaries.

4.4.2. Detection results

The columns 9–14 report detection results in Table 2 for Mask R-CNN and the proposed method SAIS. The best average results are highlighted in bold. According to the mean values of mAP, AP_{50} , and AP_{75} , the proposed method outperforms Mask R-CNN, with improvements of 3.88%, 3.32%, and 3.30%, respectively.

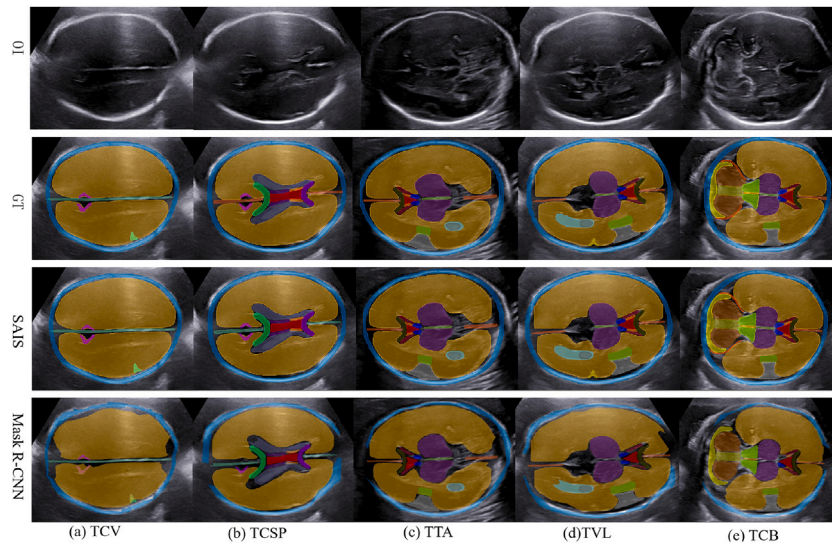
Among large structures, baseline detection performance is generally robust, yet the proposed method yields further accuracy gains. For instance, the cerebral parenchyma detection mAP increases by 4.97%, AP_{50} by 4.94%, and AP_{75} by 4.69%. Similarly, thalami show improvements of 4.47% in mAP, 3.91% in AP_{50} , and 4.16% in AP_{75} . Even for the skull, which has a very high baseline (96.87% mAP), the SAIS achieves a refined mAP of 97.77%. These consistent improvements suggest that the proposed method effectively refines localization even for large and prominent anatomical regions.

In contrast, small or morphologically complex structures, such as the tentorium cerebelli, anterior horn of the lateral ventricle, aqueduct of the midbrain, and cerebellomedullary fissure, exhibit substantial gains. Detection mAP improvements are particularly notable for the tentorium cerebelli, increasing by 5.83% (from 20.89% to 26.72%), reflecting the model's enhanced capability in localizing thin, band-like structures. Similarly, the aqueduct of the midbrain shows an improvement of 4.09% in mAP and 3.46% in AP_{75} , while the anterior horn of the lateral ventricle gains 3.16% in mAP. Experimental findings verify that the proposed structure-aware overlap metric and improved non-maximum

Table 2

Detection and segmentation results of Mask R-CNN and the proposed method SAIS on the test dataset using abbreviations. * indicates a statistically significant improvement ($p < 0.05$) of the proposed method over Mask R-CNN, based on paired Wilcoxon signed-rank tests.

No.	Abbr.	Segmentation (%)						Detection (%)					
		Mask R-CNN			SAIS			Mask R-CNN			SAIS		
		mAP	AP ₅₀	AP ₇₅	mAP	AP ₅₀	AP ₇₅	mAP	AP ₅₀	AP ₇₅	mAP	AP ₅₀	AP ₇₅
1	SK	52.45	74.30	66.37	61.97	83.37	75.75	96.87	98.75	99.05	97.77	99.85	99.85
2	FC	44.12	90.47	36.74	45.72	91.37	37.31	49.22	83.37	56.51	54.82	88.23	61.36
3	CP	87.56	97.63	92.63	88.26	98.03	93.13	83.29	92.43	88.23	88.26	96.79	92.92
4	TH	77.12	93.95	89.82	78.20	95.35	90.89	84.82	92.41	88.64	89.29	96.32	92.80
5	TV	42.72	86.91	36.12	45.22	90.01	39.19	44.82	84.06	46.91	50.19	88.83	51.29
6	PHLV	72.01	92.04	84.84	72.76	92.82	85.40	72.15	88.23	82.64	77.93	93.13	88.31
7	AHLV	33.89	73.51	47.99	53.78	91.15	62.59	54.38	86.60	67.35	57.54	89.04	70.14
8	ChP	67.18	91.93	80.01	68.38	93.63	81.44	77.83	92.41	89.60	80.08	93.69	91.32
9	SF	52.84	91.38	61.86	57.65	94.13	64.69	54.90	83.08	53.27	60.01	87.91	57.48
10	CSP	51.36	89.94	61.77	55.27	91.52	63.83	68.22	90.80	84.08	71.06	92.91	86.32
11	SP	29.71	71.02	18.56	34.27	75.02	22.55	37.55	71.72	40.93	40.44	74.28	43.21
12	STS	43.91	86.78	39.89	49.21	90.18	45.11	52.26	88.38	57.03	58.01	93.62	61.91
13	ITS	33.11	76.80	18.62	38.03	80.36	22.44	51.24	78.86	78.80	53.27	80.36	80.36
14	CF	42.62	84.00	49.07	53.48	93.01	58.81	62.58	90.54	81.39	66.32	93.29	84.26
15	POS	19.38	59.63	14.43	28.89	66.83	22.33	50.16	75.61	60.67	52.34	77.48	62.16
16	CC	58.94	85.77	79.19	68.57	94.83	87.94	66.81	88.85	85.06	69.37	90.49	86.84
17	INS	53.12	89.57	56.25	55.43	91.61	58.81	57.08	89.10	57.65	59.24	91.02	59.08
18	IFS	29.12	57.97	38.48	32.41	59.69	40.79	42.56	57.27	57.88	45.26	59.69	59.69
19	CH	76.34	92.26	86.63	78.62	94.98	89.41	83.41	89.67	86.94	88.56	94.04	92.02
20	CV	63.90	87.10	77.34	66.95	90.96	80.96	75.53	85.90	83.20	80.54	90.02	88.21
21	MB	62.72	88.20	79.47	65.21	90.41	81.64	71.80	84.93	81.13	76.69	89.47	85.04
22	TC	28.10	74.70	17.14	35.15	80.77	24.01	20.89	63.68	6.26	26.72	69.35	11.81
23	AM	33.08	68.57	35.08	40.79	75.61	41.61	34.77	65.98	27.17	38.86	69.35	30.63
24	CM	68.08	91.18	79.92	71.43	94.78	83.52	67.93	88.35	74.87	71.05	90.79	77.15
25	CMF	11.18	40.86	18.46	16.37	46.22	23.96	24.02	46.74	18.27	28.50	50.79	21.91
Avg.		49.38	81.46	54.67	54.48*	85.87*	59.12*	59.40	82.31	66.14	63.28*	85.63*	69.44*

**Fig. 6.** Segmentation results of Mask R-CNN and SAIS on fetal brain ultrasound images.

suppression algorithm significantly enhance detection accuracy for small and irregular structures.

Fig. 7 shows the original image (OI), ground truth (GT), Mask R-CNN, and the proposed SAIS detection results on fetal brain ultrasound images. The first row presents the original 2D fetal brain ultrasound images, the second row shows manually annotated labels, and the third and fourth rows show detection bounding boxes by the proposed model and Mask R-CNN, respectively. Figures (a)–(e) correspond to the TCV, TCSP, TTA, TVL, and TCB planes. As illustrated in the figure, differences between Mask R-CNN and SAIS are most evident across the five anatomical planes. Compared with the GT annotations, Mask R-CNN tends to

generate bounding boxes for structures such as FC, SP, CSP, and TV with noticeable spatial offsets. On the TVL plane, small structures, including AHLV, PHLV, and ChP, are occasionally missed or localized with imprecise bounding extents.

In contrast, SAIS shows closer agreement with the GT annotations across all planes. The predicted bounding boxes exhibit improved spatial alignment for large structures on the TCB plane, as well as more consistent localization of midline and ventricular structures on the TCSP, TTA, and TVL planes. On the TCV plane, SAIS also provides more stable detection of posterior fossa structures with reduced localization variability. Overall, SAIS achieves more accurate and consistent detection

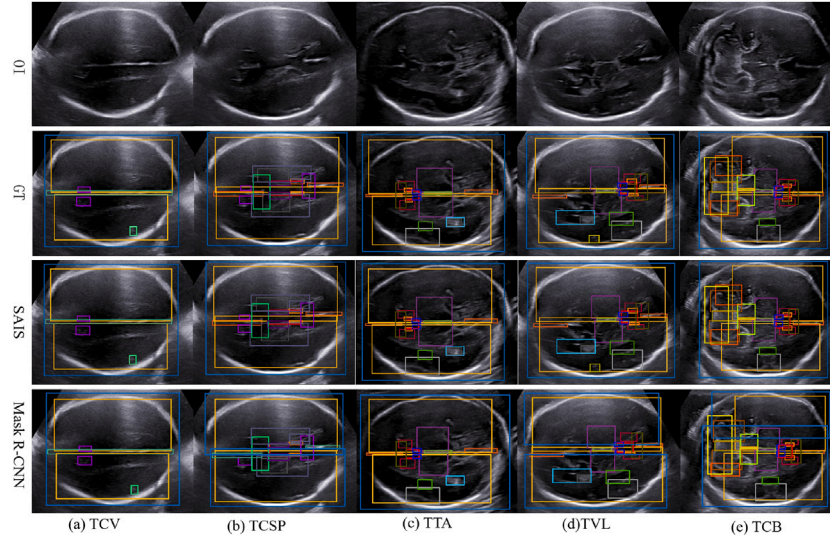


Fig. 7. Detection results of Mask R-CNN and the proposed method on fetal brain ultrasound images.

results across different anatomical planes, which is consistent with the quantitative evaluation.

4.5. Comparisons with other methods

To rigorously evaluate the effectiveness of the proposed SAIS framework, we conducted a comprehensive quantitative comparison against several state-of-the-art instance segmentation approaches, including YOLACT [45], MEInst [41], TensorMask [46], EmbedMask [47], and BlendMask [39].

The evaluation primarily focused on the mean average precision for both object detection and instance segmentation across all 25 anatomical structures, the statistical significance is assessed using paired Wilcoxon signed-rank tests. As detailed in the detection results (Table 3), our method achieved a superior average detection mAP of 63.28%, surpassing the top-performing anchor-free method, BlendMask (62.35%), as well as other advanced approaches such as EmbedMask (61.45%) and TensorMask (60.50%). These results indicate that SAIS not only generates more accurate bounding boxes for localization but also learns discriminative feature representations that are robust to the scale variations inherent in fetal ultrasound images.

In the more challenging instance segmentation task, which requires pixel-level boundary accuracy, the advantages of SAIS are even more pronounced. As shown in Table 4, it achieved an average segmentation mAP of 54.48%, outperforming the strongest competitor, BlendMask (53.11%), by 1.37%. This consistent superiority across both detection and segmentation metrics suggests that the architectural innovations in SAIS effectively capture fine-grained morphological details often missed by other state-of-the-art methods. While real-time approaches such as YOLACT prioritize speed, they suffer from substantial accuracy loss (42.74% mAP).

The gains are particularly notable in structures traditionally difficult to segment due to complex shapes, small sizes, or low contrast. For example, in the segmentation of the anterior horn of the lateral ventricle, which often exhibits poor contrast, SAIS achieved an impressive mAP of 53.78%, substantially outperforming BlendMask (36.45%). Similarly, for the cerebellomedullary fissure, one of the smallest and most challenging structures, SAIS improved the mAP to 16.37%, compared to 12.03% for the best competing method. Even for relatively distinct structures such as the skull and cerebral parenchyma, SAIS further enhanced performance, achieving 61.97% mAP for the skull versus 56.42% for BlendMask. Collectively, these results validate that SAIS effectively handles the high intra-class variability and low inter-class contrast typical of

fetal ultrasound images, providing a reliable foundation for automated biometric analysis.

Fig. 8 shows the segmentation results of the SAIS, BlendMask, EmbedMask, TensorMask, MEInst and YOLACT for 25 fetal brain anatomical structures. Figures (a)–(e) correspond to the TCV, TCSP, TTA, TVL, and TCB planes. As illustrated in the figure, several comparative methods exhibit discontinuities or partial omissions when segmenting thin and elongated anatomical structures, such as FC, SP, and CF. Small ventricular structures, including CSP and TV, are occasionally fragmented, while irregular boundaries and shape inconsistencies are observed in cortical sulci such as SF.

In comparison, SAIS yields more continuous and complete delineation across both large and small anatomical regions. Clearer boundaries are observed for major structures, including SK and CP, as well as for fine-grained regions such as SP, CSP, and the ventricular components AHLV and PHLV. Posterior fossa structures, including CH, CV, CM, and CMF, also exhibit improved spatial consistency with fewer spurious regions. Overall, our method achieves superior segmentation results.

Additionally, the segmentation performance on the fetal brain ultrasound images test set is quantitatively evaluated using mean Intersection over Union (mIoU) and mean Dice Similarity Coefficient (mDSC). Table 5 presents the quantitative comparison between the proposed SAIS and the comparative methods.

As shown in the table, SAIS achieves an mIoU of 82.72% and an mDSC of 86.01%, outperforming all the compared methods on both evaluation metrics. BlendMask yields the second-best performance, with an mIoU of 81.59% and an mDSC of 84.82%. Consequently, SAIS achieves absolute gains of 1.13% in mIoU and 1.19% in mDSC. Compared with other methods including YOLACT, MEInst, and Mask R-CNN, SAIS consistently exhibits higher quantitative segmentation accuracy on the test set across both metrics.

4.6. Ablation studies

To investigate the contributions of individual components in the proposed SAIS framework, we performed a stepwise ablation study on the fetal brain ultrasound dataset. Table 6 summarizes the results. ID 1 corresponds to the standard Mask R-CNN baseline, and subsequent IDs incrementally incorporate SAOM-NMS, FGSH, SALR, and DAGA.

Incorporating the structure-aware overlap metric and improved non-maximum suppression (SAOM-NMS) primarily targets small structure detection. Detection performance improves markedly compared with the

Table 3
Comparison of detection results (mAP %) for all 25 anatomical structures. * indicates a statistically significant improvement ($p < 0.05$) of the proposed method over other methods, based on paired Wilcoxon signed-rank tests.

Method	SK	FC	CP	TH	TV	PHLV	AHLV	ChP	SF	CSP	SP	STS	ITS	CF	POS	CC	INS	IFS	CH	CV	MB	TC	AM	CM	CMF	Avg.
YOLOACT	83.85	42.60	72.09	73.41	38.79	62.45	47.07	67.36	47.52	59.04	32.50	45.23	44.35	54.16	43.41	57.82	49.40	36.84	72.19	65.37	62.14	18.08	30.10	58.79	20.79	51.41
MEInst	89.54	45.50	76.99	78.40	41.43	66.69	50.27	71.94	50.75	63.06	34.71	48.31	47.36	57.85	46.37	61.76	52.76	39.34	77.10	69.81	66.37	19.31	32.14	62.79	22.20	54.91
TensorMask	97.05	50.80	84.50	86.10	46.50	74.00	55.10	78.40	56.50	69.10	38.50	54.10	51.75	63.50	50.70	67.45	57.60	43.25	84.70	76.70	73.00	22.35	35.80	68.70	25.10	60.50
EmbedMask	97.15	51.50	85.80	87.20	47.80	75.20	55.90	78.95	57.70	69.75	39.15	55.40	52.25	64.40	51.25	68.10	58.15	43.90	85.95	78.00	74.20	23.80	36.80	69.50	26.25	61.45
BlendMask	97.35	53.10	87.10	88.30	49.10	76.50	56.70	79.50	58.90	70.40	39.80	56.70	52.75	65.30	51.80	68.75	58.70	44.60	87.25	79.30	75.45	25.25	37.85	70.25	27.35	62.35
SAIS	97.77	54.82	88.26	89.29	50.19	77.93	57.54	80.08	60.01	71.06	40.44	58.01	53.27	66.32	52.34	69.37	59.24	45.26	88.56	80.54	76.69	26.72	38.86	71.05	28.50	63.28*

Table 4
Comparison of segmentation results (mAP %) for all 25 anatomical structures. * indicates a statistically significant improvement ($p < 0.05$) of the proposed method over other methods, based on paired Wilcoxon signed-rank tests.

Method	SK	FC	CP	TH	TV	PHLV	AHLV	ChP	SF	CSP	SP	STS	ITS	CF	POS	CC	INS	IFS	CH	CV	MB	TC	AM	CM	CMF	Avg.
YOLOACT	45.40	38.19	75.78	66.75	36.97	62.32	29.33	58.14	45.73	44.45	25.71	38.00	28.66	36.89	16.77	51.01	45.97	25.20	66.07	55.31	54.28	24.32	28.63	58.92	9.68	42.74
MEInst	48.48	40.78	80.94	71.29	39.49	66.57	31.33	62.10	48.85	47.48	27.46	40.59	30.61	39.40	17.91	54.48	49.10	26.92	70.57	59.07	57.98	25.98	30.58	62.93	10.33	45.65
TensorMask	54.80	44.45	87.75	77.40	43.50	72.20	35.41	67.48	54.40	52.60	31.04	45.88	34.59	44.53	20.25	61.58	55.50	30.42	79.76	66.76	65.53	29.36	34.56	71.13	11.68	51.59
EmbedMask	55.39	44.85	87.92	77.65	44.10	72.35	35.79	67.78	55.50	53.50	31.37	46.37	34.96	45.01	20.47	62.24	56.09	30.75	80.62	67.48	66.23	29.67	34.93	71.89	11.81	52.15
BlendMask	56.42	45.25	88.08	77.95	44.80	72.55	36.45	68.08	56.60	54.40	31.96	47.23	35.61	45.84	20.84	63.39	57.14	31.32	82.11	68.73	67.46	30.22	35.58	73.23	12.03	53.11
SAIS	61.97	45.72	88.26	78.20	45.22	72.76	53.78	68.38	57.65	55.27	34.27	49.21	38.03	53.48	28.89	68.57	55.43	32.41	78.62	66.95	65.21	35.15	40.79	71.43	16.37	54.48*



Fig. 8. Segmentation results of the proposed method and comparison methods for 25 fetal brain structures.

Table 5

Quantitative comparison of segmentation performance on the fetal brain ultrasound image test set.

Method	mIoU (%)	mDSC (%)
YOLACT	66.46	70.26
MEInst	70.99	73.91
Mask R-CNN	74.79	78.86
TensorMask	77.23	82.39
EmbedMask	79.10	83.79
BlendMask	81.59	84.82
SAIS	82.72	86.01

baseline: mAP increases from 59.40% to 61.20%, AP_{50} from 82.31% to 83.50%, and AP_{75} from 66.14% to 67.50%. Segmentation also benefits due to more accurate RoI proposals, with mAP rising from 49.38% to 51.20%, AP_{50} from 81.46% to 82.80%, and AP_{75} from 54.67% to 56.10%. These results indicate that SAOM-NMS effectively reduces missed detections of small and thin fetal brain structures.

Adding the FGSH mainly enhances segmentation performance by expanding the receptive field and preserving fine anatomical details. Segmentation mAP improves from 51.20% to 52.78%, AP_{50} from 82.80% to 83.51%, and AP_{75} from 56.10% to 57.22% relative to ID 2. Detection also shows a modest improvement (mAP from 61.20% to 61.80%), suggesting that feature refinement in RoI branches benefits both tasks.

The SALR enforces structural consistency and smoothness in each predicted mask. Compared with ID 3, segmentation mAP increases

from 52.78% to 53.50%, AP_{50} from 83.51% to 84.20%, and AP_{75} from 57.22% to 57.90% (approximately +0.7%). Detection performance also improves slightly, with mAP rising from 61.80% to 62.50%. These results demonstrate SALR's effectiveness in refining segmentation for morphologically complex structures.

Finally, the DAGA introduces synthetic fetal brain ultrasound images to increase structural diversity. Its contribution is small but consistent: detection mAP increases from 62.50% to 63.28%, AP_{50} from 84.80% to 85.63%, and AP_{75} from 68.90% to 69.44%; segmentation mAP improves from 53.50% to 54.48%, AP_{50} from 84.20% to 85.87%, and AP_{75} from 57.90% to 59.12%. This demonstrates that DAGA provides minor but reliable gains for both detection and segmentation.

4.7. Hyperparameter sensitivity analysis

Given that the proposed structure-aware overlap metric is incorporated into the RPN to prevent small yet clinically important targets from being suppressed as background, the threshold used in the SAOM-based non-maximum suppression becomes a critical hyperparameter. To determine the optimal value, we performed a sensitivity analysis by varying the NMS threshold from 0.5 to 0.9. As shown in Table 7, thresholds in the range of 0.5-0.6 cause the segmentation mAP to drop to 61.12%, indicating that overly aggressive suppression removes valid proposals. In contrast, thresholds of 0.8-0.9 insufficiently eliminate redundancy, resulting in an accumulation of false positives that deteriorates classification performance. The model exhibits the best results at a threshold of 0.7, achieving a detection mAP of 54.48% and a segmentation mAP of 63.28%. These findings confirm that a threshold of 0.7 provides the

Table 6
The results of the ablation study for the proposed SAIS.

ID	SAOM-NMS	FGSH	SALR	DAGA	Detection (%)			Segmentation (%)		
					mAP	AP ₅₀	AP ₇₅	mAP	AP ₅₀	AP ₇₅
1					59.40	82.31	66.14	49.38	81.46	54.67
2	✓				61.20	83.50	67.50	51.20	82.80	56.10
3	✓	✓			61.80	84.10	68.19	52.78	83.51	57.22
4	✓	✓	✓		62.50	84.80	68.90	53.50	84.20	57.90
5	✓	✓	✓	✓	63.28	85.63	69.44	54.48	85.87	59.12

Table 7
Sensitivity analysis of the SAOM-NMS threshold.

SAOM-NMS Threshold	Detection (%)			Segmentation (%)		
	mAP	AP ₅₀	AP ₇₅	mAP	AP ₅₀	AP ₇₅
0.5	52.15	83.20	56.50	61.12	83.45	67.20
0.6	53.62	84.95	58.05	62.45	84.80	68.55
0.7	54.48	85.87	59.12	63.28	85.63	69.44
0.8	53.90	85.10	58.40	62.75	85.15	68.80
0.9	52.85	84.35	57.25	61.54	84.05	67.50

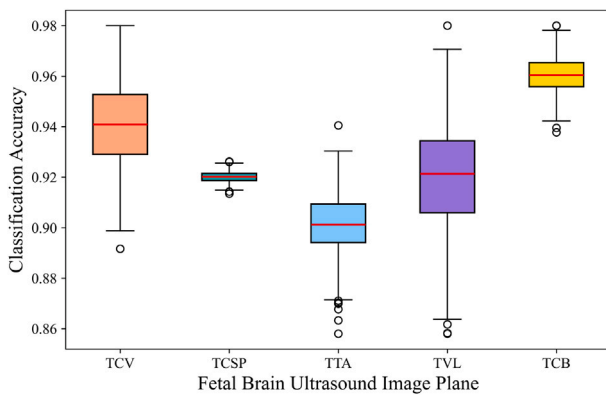


Fig. 9. Box plots of the classification accuracy on the real test dataset using the TSTR protocol. The model was trained exclusively on DAGA-generated synthetic images.

optimal balance between removing redundant detections and preserving the recall of clinically meaningful anatomical structures.

4.8. Validation of synthetic fetal brain images for backbone pre-training

The primary goal of our DAGA strategy is to enhance the segmentation model's generalization, specifically to address dynamic anatomical changes during fetal development. To achieve this, we utilize synthetic fetal brain ultrasound images to pre-train the ResNet-101-FPN backbone.

To rigorously assess this, we adopted the "Train on Synthetic, Test on Real" (TSTR) protocol. We formulated the pre-training task as the classification of five standard fetal brain planes (TCV, TCSP, TTA, TVL, TCB). Using a ResNet-101-FPN architecture identical to the backbone in our SAIS framework, the model was trained exclusively on DAGA-generated synthetic images and evaluated on real data. This setup directly verifies whether features learned from synthetic data can effectively transfer to real clinical scenarios.

As illustrated in Fig. 9. The results indicate strong transferability, with median accuracies exceeding 90% across all five anatomical planes. The TCB plane achieved the highest median accuracy of approximately 96%, followed closely by TCV. The TCSP plane showed the greatest stability, evidenced by its compact interquartile range and minimal prediction variance. Although TTA and TVL exhibited slightly

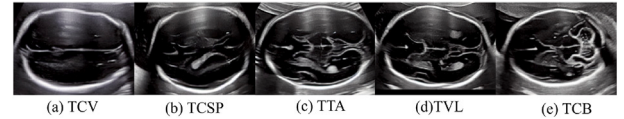


Fig. 10. Visualization of synthetic fetal brain ultrasound images generated by the DAGA strategy. These samples exhibit distinct anatomical features suitable for model pre-training.

broader distributions, their consistently high accuracies further confirm that the anatomical features learned solely from DAGA-generated images are semantically reliable and can be effectively transferred to real clinical data, validating their suitability for pre-training the segmentation backbone.

The Fig. 10 displays representative synthetic ultrasound images for the five standard fetal brain planes. Visually, the generated images exhibit clear and distinct anatomical structures, such as the cerebellum in the TCB plane and the cavum septi pellucidi in the TCSP plane. The high fidelity of these anatomical features suggests that the generator has successfully captured the semantic characteristics.

5. Discussion

In summary, the proposed structure-aware instance segmentation framework provides a comprehensive solution for fetal brain ultrasound analysis, supporting the fine-grained segmentation of 25 anatomical structures across five standard planes. The limitations of traditional detection metrics are effectively addressed through the introduction of SAOM and the improved SAOM-NMS algorithm, which substantially reduce false negatives and enhance the reliability of detecting small fetal brain structures. In addition, the fine-grained segmentation head enlarges the receptive field and strengthens feature representation, ensuring that the predicted masks maintain high accuracy. The integration of Stable Diffusion with Low-Rank Adaptation further mitigates data scarcity by generating anatomically diverse fetal brain ultrasound images, thereby improving model robustness and generalization.

Comprehensive experiments demonstrate that the proposed framework consistently outperforms state-of-the-art methods in both detection and segmentation accuracy, especially for anatomically challenging small structures. Although the method achieves promising results, it is still limited by insufficient training data and class imbalance. In the future, constructing multi-center datasets could further improve the model's robustness and clinical utility, enabling broader application of artificial intelligence in fetal brain ultrasound image analysis.

The proposed framework is built upon a two-stage instance segmentation architecture based on Mask R-CNN, whose inference-time computational complexity is comparable to that of standard Mask R-CNN models. Although the proposed structure-aware strategies enhance segmentation accuracy and spatial consistency, the two-stage inference paradigm may limit real-time deployment on resource-constrained clinical devices. The additional modules mainly affect the training stage, while the inference pipeline remains unchanged in structure.

Moreover, the domain-adaptive generative augmentation strategy is applied only during training and does not introduce additional

inference-time overhead, but it increases the overall training complexity. Consequently, the current method is more suitable for offline analysis or deployment on GPU-equipped clinical workstations.

All experiments are conducted on a single-center fetal brain ultrasound dataset, and the proposed method has not yet been validated on data from other clinical centers. Although the framework is tailored for fetal brain ultrasound images with relatively stable anatomical configurations, its robustness to variations in imaging devices, acquisition protocols, and operator-dependent factors remains to be investigated.

CRedit authorship contribution statement

Laifa Ma: Validation, Software, Methodology, Conceptualization. **Kenli Li:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **Zheng Xiao:** Writing – review & editing, Software. **Guanghua Tan:** Validation, Software, Formal analysis. **Huaxuan Wen:** Validation, Resources, Investigation, Data curation. **Shengli Li:** Writing – review & editing, Resources, Project administration, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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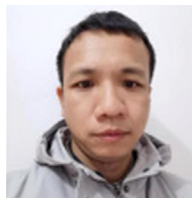
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